Generative Modelling with Stochastic Differential Equations

Exercises, chapter 1

Sampling in high dimension

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1 Sampling in high dimension

Exercise 1.1 (Bayesian logistic regression)**.** Let $(a_i, y_i) \in \mathbb{R}^n \times \{-1, +1\}$, $i = 1, \dots, N$, and $\lambda > 0$, and consider the potential

$$U(\theta) := \sum_{i=1}^{N} \ell(y_i\, a_i^\top \theta) + \frac{\lambda}{2}|\theta|^2, \qquad \ell(s) := \log(1 + e^{-s}), \qquad \theta \in \mathbb{R}^n.$$

This is the potential of the posterior in Bayesian logistic regression (Example 1.1).

(a) Compute ∇U and $\nabla^2 U$, and show that $0 < \ell'' \le \frac{1}{4}$.

(b) Deduce that $\alpha\, \mathbb{I}_n \preceq \nabla^2 U(\theta) \preceq \beta\, \mathbb{I}_n$ for all θ, with $\alpha = \lambda$ and $\beta = \lambda + \frac{1}{4}\lambda_{\max}\big(\sum_i a_i a_i^\top\big) \le \lambda + \frac{1}{4}\sum_i |a_i|^2$.

(c) Show that $Z = \int e^{-U}$ is finite, and that U has a unique minimiser $\theta^\star$ (the maximum a posteriori estimator).

Exercise 1.2 (Kernel density estimation of a Lipschitz density)**.** Let f be a probability density on $\mathbb{R}^n$ which is L-Lipschitz and bounded by M, and let $x_1, \dots, x_N$ be i.i.d. with density f. Let $K \ge 0$ be a probability density on $\mathbb{R}^n$ with $m_1 := \int |u| K(u)\, \mathrm{d}u < \infty$ and $\|K\|_2^2 := \int K(u)^2\, \mathrm{d}u < \infty$, and let $\hat{f}_h(x) = \frac{1}{Nh^n} \sum_{i=1}^{N} K\big(\frac{x - x_i}{h}\big)$ be the kernel density estimator with bandwidth $h > 0$.

(a) Show that $|\mathbb{E}\hat{f}_h(x) - f(x)| \le L m_1 h$ for every $x \in \mathbb{R}^n$.

(b) Show that $\operatorname{Var} \hat{f}_h(x) \le M\|K\|_2^2/(Nh^n)$ for every $x \in \mathbb{R}^n$.

(c) Deduce that

$$\mathbb{E}\big(\hat{f}_h(x) - f(x)\big)^2 \le L^2 m_1^2\, h^2 + \frac{M\|K\|_2^2}{Nh^n},$$

and that with $h = N^{-1/(n+2)}$ the mean squared error is at most $C N^{-2/(n+2)}$, where $C := L^2 m_1^2 + M\|K\|_2^2$. Conclude that it is at most ε once $N \ge (C/\varepsilon)^{(n+2)/2}$.

Rejection sampling. Exercises 1.3 and 1.4 use the following setting. The target is a probability density $\mu = \tilde{\mu}/Z_\mu$ on $\mathbb{R}^n$, known only through an unnormalized density $\tilde{\mu} \ge 0$ with $Z_\mu := \int \tilde{\mu} \in (0, \infty)$. The proposal is a density $\nu = \tilde{\nu}/Z_\nu$ that we can sample from, where $Z_\nu := \int \tilde{\nu} < \infty$ and $\tilde{\nu} \ge \tilde{\mu}$ everywhere ($\tilde{\nu}$ is an *upper envelope* of $\tilde{\mu}$). Rejection sampling draws $X \sim \nu$ and an independent $U \sim \mathrm{Unif}[0, 1]$, and accepts X if $U \le \tilde{\mu}(X)/\tilde{\nu}(X)$. Otherwise it repeats with a new independent pair. The output is the first accepted proposal. The *acceptance probability* is the probability that a single proposal is accepted.

Exercise 1.3 (Rejection sampling of a Gaussian)**.** Let the target be $\mu = \mathcal{N}(0, \mathbb{I}_n)$ with $\tilde{\mu}(x) = e^{-|x|^2/2}$, and the proposal $\nu = \mathcal{N}(0, s^2 \mathbb{I}_n)$ for some $s > 0$, with envelopes of the form $\tilde{\nu}_c(x) = c\, e^{-|x|^2/(2s^2)}$, $c > 0$.

(a) Show that an envelope $\tilde{\nu}_c \ge \tilde{\mu}$ exists if and only if $s \ge 1$, and that the smallest valid c is then $c = 1$.

(b) Show that the acceptance probability with $c = 1$ is s^{-n}.

(c) With $s = 1.01$, how many proposals are needed on average when $n = 1000$?

(d) Now let the target be $\mathcal{N}(m, \mathbb{I}_n)$ and keep the proposal $\mathcal{N}(0, s^2\mathbb{I}_n)$, $s > 1$. Show that the best acceptance probability is $s^{-n} \exp\left(-\frac{|m|^2}{2(s^2-1)}\right)$.

Exercise 1.4 (Volume of the unit ball)**.** Let $\mathcal{B}_2^n = \{x \in \mathbb{R}^n : |x| \le 1\}$ and let $\omega_n := |\mathcal{B}_2^n|$ be its volume.

(a) Using $\int_{\mathbb{R}} e^{-t^2} \mathrm{d}t = \sqrt{\pi}$ and the layer-cake formula $\int_{\mathbb{R}^n} \varphi \, \mathrm{d}x = \int_0^\infty |\{\varphi > t\}| \, \mathrm{d}t$ for $\varphi \ge 0$, compute $\int_{\mathbb{R}^n} e^{-|x|^2} \mathrm{d}x$ in two ways and deduce

$$\omega_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}.$$

(b) Show that $\omega_{n+2} = \frac{2\pi}{n+2} \, \omega_n$, and deduce that $\omega_n < 1$ if and only if $n \ge 13$. Consider rejection sampling of the uniform distribution on $\mathcal{B}_2^n$ with $\tilde{\mu} = \mathbf{1}_{\mathcal{B}_2^n}$ and envelope $\tilde{\nu} = \mathbf{1}_{[-1,1]^n}$. Show that its acceptance probability is $\omega_n / 2^n$ and conclude that it is smaller than 2^{-n} if and only if $n \ge 13$.

(c) Using Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x} \, (x/e)^x$, show that $\omega_n 2^{-n} \sim \frac{1}{\sqrt{\pi n}} \left(\frac{\pi e}{2n}\right)^{n/2}$.

Exercise 1.5 (Maximum likelihood is KL minimisation)**.** Recall the Kullback–Leibler divergence $\mathrm{KL}(\mu \| g) := \int \mu \log(\mu/g)$ (a sum on a finite set), with $\mathrm{KL}(\mu\|g) = +\infty$ if μ is not absolutely continuous with respect to g, and the maximum likelihood estimator $\arg\max_{g \in \mathcal{G}} \frac{1}{N} \sum_{i=1}^N \log g(x_i)$ over a family $\mathcal{G}$ of densities.

(a) Let μ and g be probability densities on $\mathbb{R}^n$ with $\mathbb{E}_\mu |\log \mu| < \infty$ and $\mathbb{E}_\mu |\log g| < \infty$. Show that $\mathbb{E}_{X \sim \mu} \log g(X) = \mathbb{E}_\mu \log \mu - \mathrm{KL}(\mu\|g)$, and that $\mathrm{KL}(\mu\|g) \ge 0$ with equality iff $g = \mu$ a.e. Deduce that, over a family $\mathcal{G}$, $\arg\max_{g \in \mathcal{G}} \mathbb{E}_\mu \log g = \arg\min_{g \in \mathcal{G}} \mathrm{KL}(\mu\|g)$, which contains μ if $\mu \in \mathcal{G}$.

(b) Let μ be a distribution on a finite set and $\hat{\mu}_N = \frac{1}{N} \sum_i \delta_{x_i}$ the empirical measure. Show that the MLE over a family $\mathcal{G}$ of probability mass functions is exactly the minimiser of $g \mapsto \mathrm{KL}(\hat{\mu}_N \| g)$ over $\mathcal{G}$. Why does this fail for densities on $\mathbb{R}^n$?

(c) Let $\mathcal{G} = \{\mathcal{N}(m, \Sigma) : m \in \mathbb{R}^n, \Sigma \succ 0\}$ and let μ be a probability density on $\mathbb{R}^n$ with $\mathbb{E}_\mu |\log \mu| < \infty$, finite second moments, mean m_μ and covariance $\Sigma_\mu \succ 0$. Show that $\arg\min_{g \in \mathcal{G}} \mathrm{KL}(\mu\|g) = \mathcal{N}(m_\mu, \Sigma_\mu)$. Deduce that, when the empirical covariance of $x_1, \dots, x_N$ is positive definite, the Gaussian MLE is the Gaussian with the empirical mean and covariance.